Vortex
Flow Meters

# VN2000 Flow Meters

## Compact Insertion Meter

## DESCRIPTION

The VN2000 Compact Insertion Vortex Flow Meter measures the flow of liquids over a large flow range. The meter includes a mounting assembly that welds to the pipe and an alignment pin to orient the meter properly during installation. The meter is designed for specific pipe sizes and includes parts for installation.

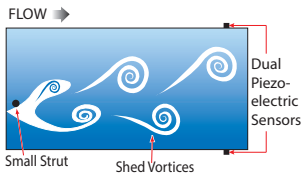
The vortex sensing element is CNC machined out of one piece of solid stainless steel and engineered to stand up to abusive environments inside and outside the pipe. The dual ceramic piezoelectric sensors are bonded inside the vortex element, which is press-fit to the stainless steel insertion bar and completely welded together. There are no internal O-rings or seals of any kind and absolutely no leak paths into the sensors or electronics. The sensors do not touch the process fluid, which gives them an almost unlimited life span.

## BENEFITS

- Measure volume or Btu of liquids
- Lower profile insertion meter for pipes 2–24 in.
- Simplify installation with an insertion meter and pipe mounting assembly designed for the specific pipe size
- Reduced noise interference with dual piezoelectric sensors and filtering
- Standard model handles process temperatures up to 400 °F (204 °C)
- Reduced maintenance due to:
    - Heavy duty welded stainless steel construction
    - O-ring-free element
    - No moving parts
    - Piezoelectric and temperature sensors never touch process fluid

## OPERATION

An everyday example of a vortex shedding phenomenon is a flag waving in the breeze: the flag waves due to the vortices shed by air moving across the flagpole. Within the flow meter, as flowing fluid moves across the tiny strut or "shedder bar", vortices are shed on a smaller scale. The frequency of the vortices shedding is proportional to the fluid velocity.






Through the use of an internal RTD, the flow meter software compensates for changes in temperature to achieve an accurate mass flow measurement.

With the addition of a second external RTD, the meter can measure the energy transfer across a heat exchanger.

## APPLICATIONS

The unique design of the VN2000 meter can be used in liquid lines.

## INSTALLATION

The VN2000 compact meter includes a mounting assembly with alignment pin and an arrow machined into the mounting assembly that shows the direction of flow.

After depressurizing the pipe and welding the mounting assembly onto a welded outlet, insert the VN2000 compact meter into the mounting assembly and align the meter so the pin can be inserted into the mounting assembly. When the pin is inserted, you are ensured that the meter is inserted to the proper depth and is aligned with the flow. The pin also acts as a secondary method to prevent the meter from coming out of the pipe once the pipe is pressurized.

## SIZING SOFTWARE

The Vortex Sizing Application software determines the precise scaling factor for your application based on:

- Fluid type
- Min. and max. operating flow rate
- Operating temperature
- Operating pressure

Badger Meter

VRX-DS-02212-EN-12 (May 2026)

Product Data Sheet

## SPECIFICATIONS

| | Uncertainty | |
|---|---|---|
| | **Volumetric Flow** | **Repeatability** |
| **Liquids** | ±1.0% of reading | ±0.25% of reading |
| | **Mass or Heat Flow** | **Repeatability** |
| **Liquids** | ±1.0% of reading | ±0.25% of reading |
| **Velocity – Liquid** | 1.32–32 ft/s (0.402–9 m/s) | |
| **Reynolds Range** | 10,000–7,000,000 depending on fluid density | |
| **Straight Run Pipe** | Upstream 10 diameters; downstream 5 diameters with one 90° elbow before the meter | |
| **Pipe Sizes** | 2–36 in. (50–900 mm) | |
| **Fluid Temperature** | –250 to 400 °F (–120 to 204 °C) | |
| **Maximum Pressure** | 1000 psi (68.9 bar) | |
| **Wetted Materials** | Stainless steel 304L | |
| **Hot Tap Connections** | 1-1/2 in: NPT, ASME/ANSI 150, 300 flanges; Sealing assembly: Two ethylene propylene O-rings; Optional removable extractor | |

| **Measurement Options** | **Sensors** | | |
|---|---|---|---|
| Volumetric flow rate | Dual piezo vortex sensors | | |
| Mass flow rate based on velocity and temperature | Dual piezo vortex sensors<br>RTD embedded in sensor: 100 Ohm, 3 wire | | |
| BTU/Energy | Dual piezo vortex sensors<br>RTD embedded in sensor: 100 Ohm, 3 wire<br>External 100 Ohm RTD input | | |
| Units of measure | **Units Base** | **Liquids** | **Energy Meter** |
| | English | Pounds (LB)<br>Tons (TON)<br>Cubic Feet ($FT^3$)<br>Gallons (GAL)<br>Oil Barrels (BBL) | British Thermal Units (BTU)<br>Tons (TON) |
| | Metric | Kilograms (KG)<br>Tonnes (TNN)<br>Cubic Meters ($M^3$)<br>Liters (LTR) | — |
| | Actual English | — | — |
| | Natural Metric | — | — |
| | Units of measure are used for flow rate, mass flow rate, heat/energy flow rate; volume total, mass total, heat/energy total | | |
| Measurement interval | Second, minute, hour, day | | |

### Transmitter

| | |
|---|---|
| **Display** | 2 × 16 characters reflective display, Rotatable display<br>Flow rate: 6 digits with decimal<br>Totalizer |
| **Keypad** | 4 membrane buttons |
| **Power** | 14–36V DC; loop powered with 4-20 mA option, 28V DC max. |
| **Operating Temperature** | 32 to 140 °F (0 to 60 °C)<br>5–95% relative humidity non-condensing |
| **Output** | One 4-20 mA, 10–36V max load, 24-bit resolution; Available Communication and I/O Standard option "S" |
| **Digital Output** | One min. input resistance 1000 Ohm<br>Max output frequency = 12.5 Hz<br>Opto isolator 5–24V DC<br>Available Communication and I/O Standard option "S" |
| **Communication** | EIA-485 with Modbus RTU (9600, 14400, 19200 baud) available Communication and I/O Standard option "M" or<br>BACnet MS/TP (9600, 19200, 38400, 76800 baud) available Communication and I/O Standard option "B" |
| **Enclosure** | General purpose |
| **Mounting** | Integral meter or remote pipe and wall; Up to 30 ft (10 m) from pipe |
| **Cable** | Cable jacket material: polyurethane (gray)<br>Operating temperature –13 to 185 °F (–25 to 85 °C) |

The power consumption of the 4-20 mA version is max. 5 Watts; the BACnet or Modbus version is max. 1 Watt.

# FLOW RATE TABLES

## Liquid Minimum to Maximum Flow Rates in US Gal/min *

| Pipe Line Size in. | Density 62.3 lb/ft$^3$ |
|---|---|
| 2 | 13.80–313.7 gpm |
| 3 | 30.41–691.2 gpm |
| 4 | 52.37–1190 gpm |
| 6 | 118.9–2701 gpm |
| 8 | 205.8–4678 gpm |
| 10 | 324.4–7373 gpm |
| 12 | 460.5–10466 gpm |
| 14 | 556.6–12650 gpm |
| 16 | 727.0–16523 gpm |
| 18 | 920.2–20914 gpm |
| 20 | 1143–25988 gpm |
| 22 | 1459–33160 gpm |
| 24 | 1654–37587 gpm |

* Standard conditions of 70 °F (21.11 °C)

## Liquid Minimum to Maximum Flow Rates in Liter/min *

| Pipe Line Size in. | Density 997.77 kg/m$^3$ |
|---|---|
| 50 | 52.26–1188 lpm |
| 75 | 115.1–2616 lpm |
| 100 | 198.2–4506 lpm |
| 150 | 449.9–10225 lpm |
| 200 | 779.1–17706 lpm |
| 250 | 1228–27909 lpm |
| 300 | 1743–39616 lpm |
| 350 | 2107–47885 lpm |
| 400 | 2752–62544 lpm |
| 450 | 3483–79167 lpm |
| 500 | 4328–98373 lpm |
| 550 | 5523–125523 lpm |
| 600 | 6260–142280 lpm |

* Standard conditions of 70 °F (21.11 °C)

# DIMENSIONS

## VN2000 Compact Insertion Meter

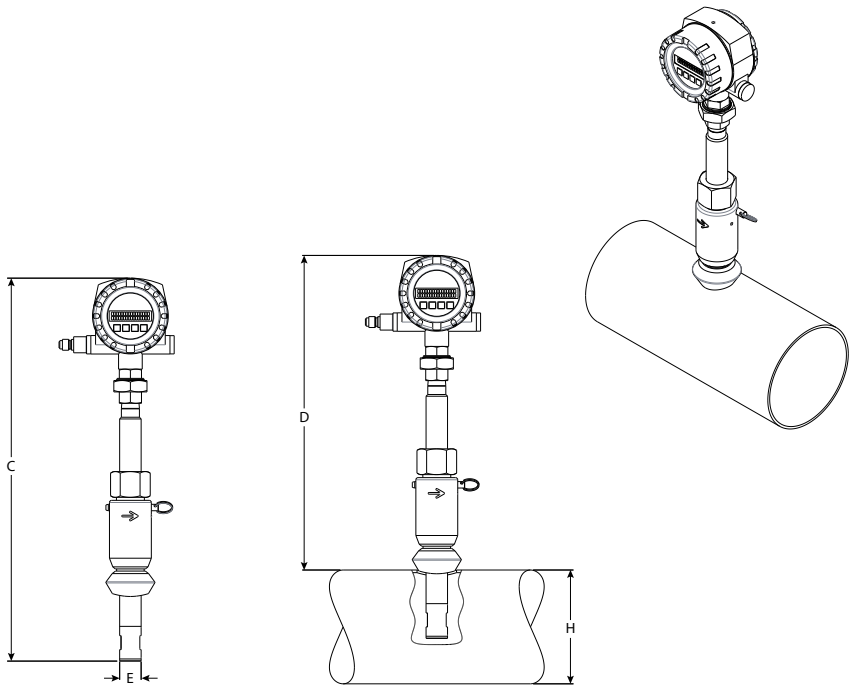



*Figure 1: Compact meter*

| H | | C | | D | | E | |
|---|---|---|---|---|---|---|---|
| in. | mm | in. | mm | in. | mm | in. | mm |
| 2.375 | 51 | 20.400 | 518.160 | 18.392 | 467.1441 | 1.25 | 32 |
| 3.5 | 76 | 20.400 | 518.160 | 17.829 | 452.8566 | 1.25 | 32 |
| 4.5 | 102 | 21.400 | 543.560 | 18.329 | 465.5566 | 1.25 | 32 |
| 5.563 | 127 | 21.400 | 543.560 | 17.798 | 452.0565 | 1.25 | 32 |
| 6.625 | 152 | 22.400 | 568.960 | 18.267 | 463.9691 | 1.25 | 32 |
| 8.625 | 203 | 23.400 | 594.360 | 18.267 | 463.9691 | 1.25 | 32 |
| 10.75 | 254 | 24.400 | 619.760 | 18.204 | 462.3816 | 1.25 | 32 |
| 12.75 | 305 | 25.400 | 645.160 | 18.204 | 462.3816 | 1.25 | 32 |
| 14 | 356 | 26.400 | 670.560 | 18.579 | 471.9066 | 1.25 | 32 |
| 16 | 406 | 27.400 | 695.960 | 18.579 | 471.9066 | 1.25 | 32 |
| 18 | 457 | 28.400 | 721.360 | 18.579 | 471.9066 | 1.25 | 32 |
| 20 | 508 | 29.400 | 746.760 | 18.579 | 471.9066 | 1.25 | 32 |
| 22 | 559 | 30.400 | 772.160 | 18.579 | 471.9066 | 1.25 | 32 |
| 24 | 610 | 31.400 | 797.56 | 18.579 | 471.9066 | 1.25 | 32 |

## VN2000 Compact Insertion Meter, Remote

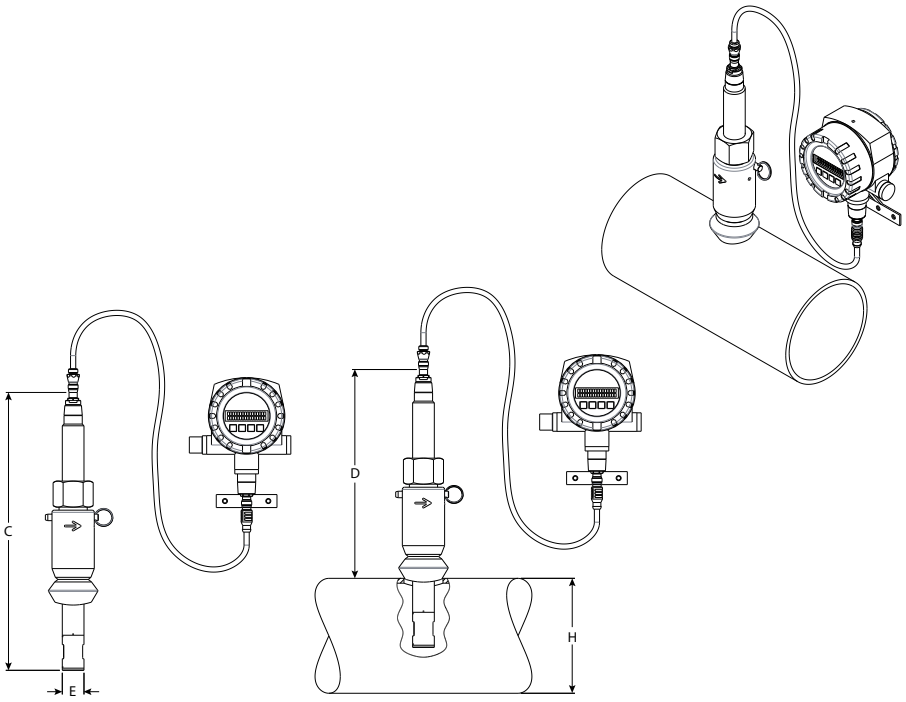



*Figure 2: Compact meter, remote*

| H | | C | | D | | E | |
|---|---|---|---|---|---|---|---|
| **in.** | **mm** | **in.** | **mm** | **in.** | **mm** | **in.** | **mm** |
| 2.375 | 51 | 14.275 | 362.585 | 12.267 | 311.569 | 1.25 | 32 |
| 3.5 | 76 | 14.275 | 362.585 | 11.704 | 297.282 | 1.25 | 32 |
| 4.5 | 102 | 15.275 | 387.985 | 12.204 | 309.982 | 1.25 | 32 |
| 5.563 | 127 | 15.275 | 387.985 | 11.673 | 296.482 | 1.25 | 32 |
| 6.625 | 152 | 16.275 | 413.385 | 12.142 | 308.394 | 1.25 | 32 |
| 8.625 | 203 | 17.275 | 438.785 | 12.142 | 308.394 | 1.25 | 32 |
| 10.75 | 254 | 18.275 | 464.185 | 12.079 | 306.807 | 1.25 | 32 |
| 12.75 | 305 | 19.275 | 489.585 | 12.079 | 306.807 | 1.25 | 32 |
| 14 | 356 | 20.275 | 514.985 | 12.454 | 316.332 | 1.25 | 32 |
| 16 | 406 | 21.275 | 540.385 | 12.454 | 316.332 | 1.25 | 32 |
| 18 | 457 | 22.275 | 565.785 | 12.454 | 316.332 | 1.25 | 32 |
| 20 | 508 | 23.275 | 591.185 | 12.454 | 316.332 | 1.25 | 32 |
| 22 | 559 | 24.275 | 616.585 | 12.454 | 316.332 | 1.25 | 32 |
| 24 | 610 | 25.275 | 641.985 | 12.454 | 316.332 | 1.25 | 32 |

## Power Plug

The multi-pole connector virtually eliminates wiring errors in the field. Simply plug the connector into the top of the enclosure and screw it down. This design saves time and money because the electronics package never comes out of the enclosure. No internal wiring is required and no connecting of hard-to-reach terminal blocks. This design also allows for a conduit connection to be attached over the power plug.

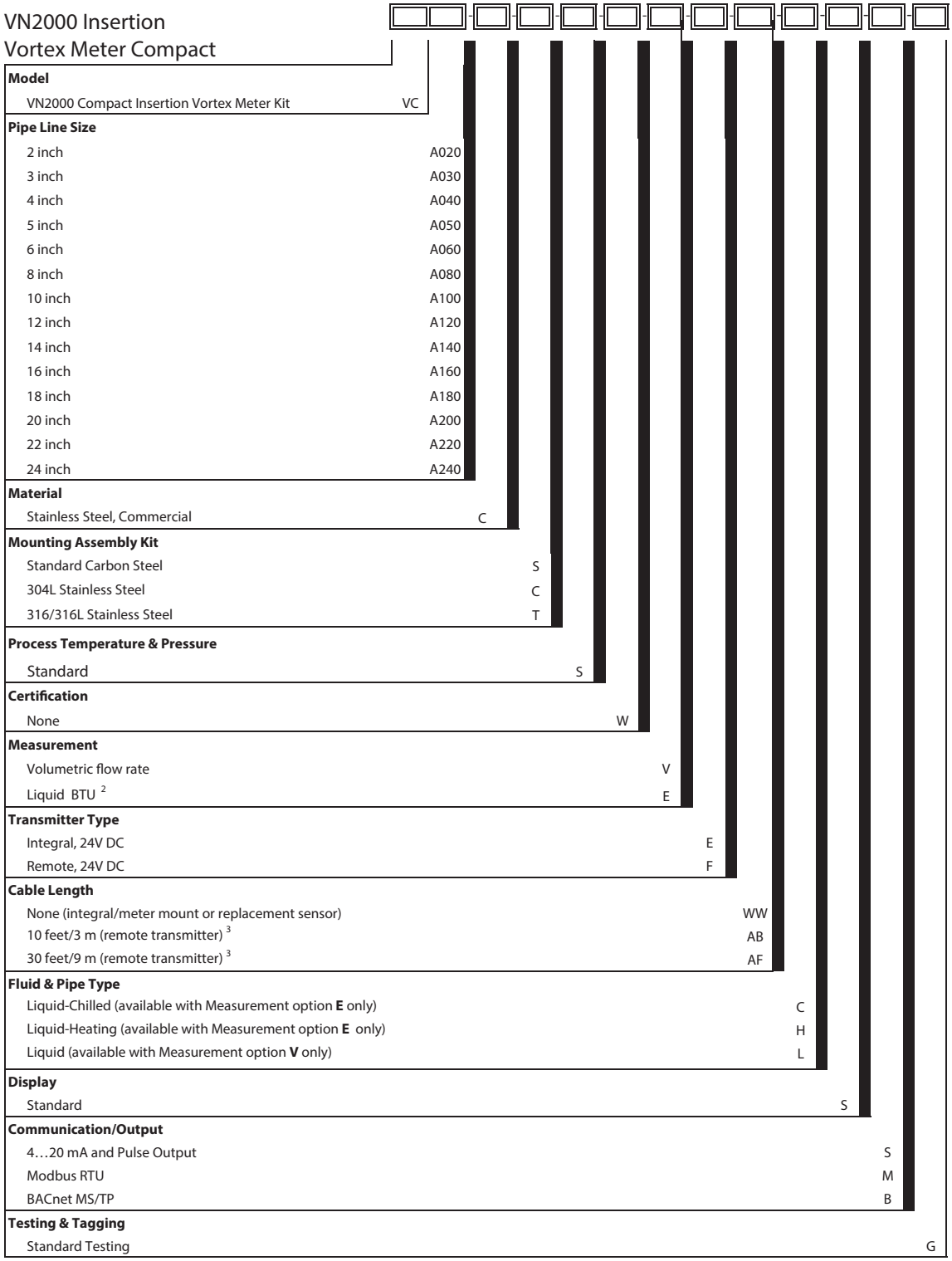
Simply plug and play!



## Optional External RTD for BTU/Energy Measurement

An insertion RTD (part number VNA-RTD) is used in conjunction with an RTD internal to the probe to measure the temperature difference across a heat exchanger. The energy transfer can be calculated by multiplying the temperature difference with the mass flow rate.

## PART NUMBER CONSTRUCTION

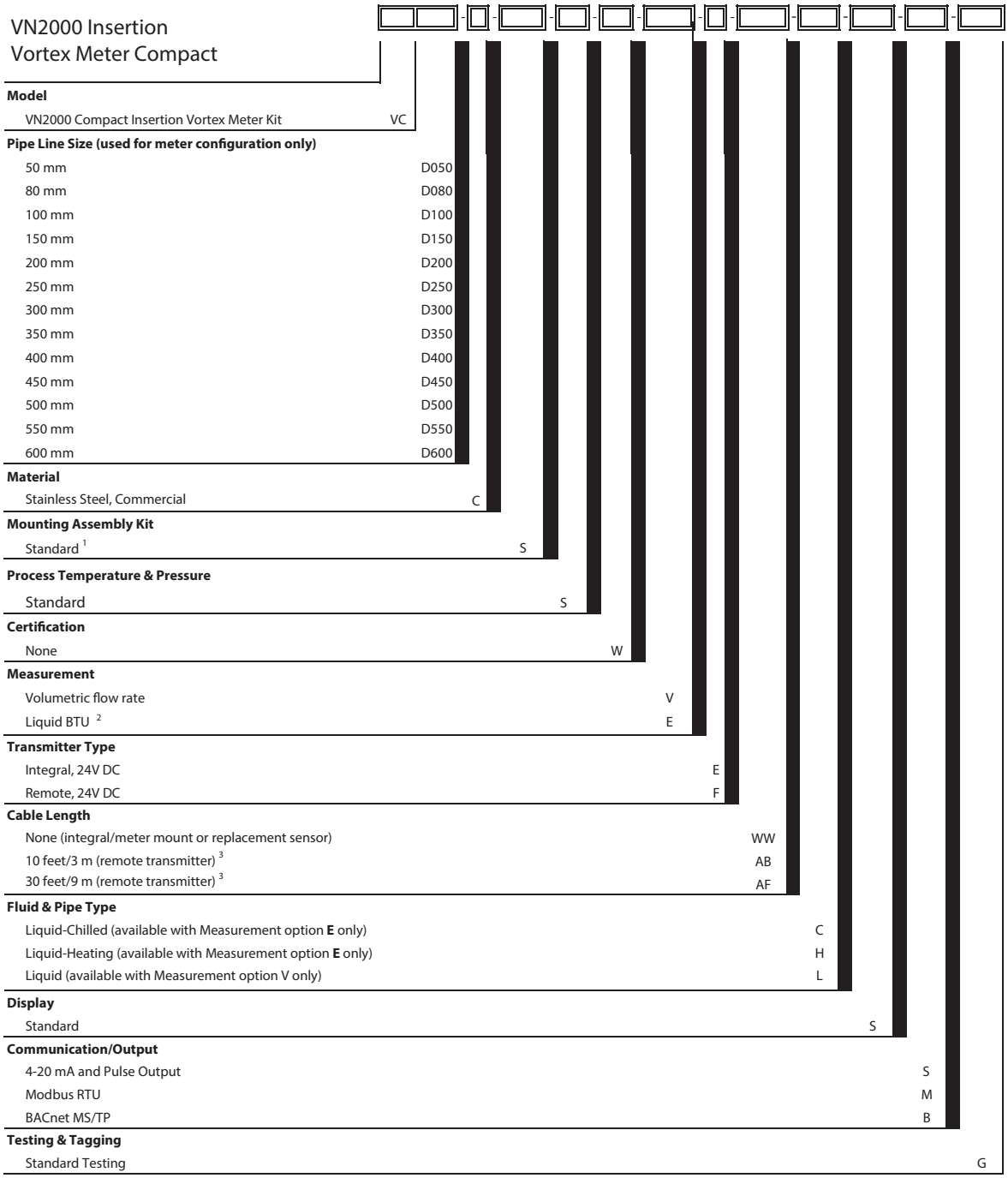



[1] Includes weldable outlet and mounting assembly with pin.

[2] One internal temperature sensor and one external temperature sensor.

[3] Specify cable length from sensor to transmitter. Transmitter power 10 foot cable included.

## Metric Pipes




1 Includes weldable outlet and mounting assembly with pin.

2 One internal temperature sensor and one external temperature sensor.

3 Specify cable length from sensor to transmitter. Transmitter power 10 foot cable included.

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